

TNSM Research Positioning and Dataset Strategy

A focused evaluation plan for 5G anomaly detection, early warning, root-cause analysis, and cross-domain generalization

CORE RECOMMENDATION

Use TelecomTS as the primary 5G domain benchmark, RCAEval as the service-management / RCA rigor benchmark, and SMD only as an out-of-domain generalization benchmark.

Target venue: IEEE Transactions on Network and Service Management (TNSM)

Executive Summary

This report recommends a deliberately narrow TNSM positioning built around four primary scope anchors and one supporting anchor. The goal is to make the paper read first as a network-and-service-management contribution and only secondarily as a machine-learning paper. The strongest primary anchors are: (1) Management Functions, (2) Service Provisioning, Reliability and Quality Assurance, (3) Enabling Technologies, and (4) Emerging Technologies and Standards. Applications and Case Studies serves as a fifth supporting dimension through realistic 5G evaluation.

Dataset strategy. TelecomTS should be the primary benchmark because it is derived from a 5G telecommunications testbed and exposes high-resolution PHY/MAC/network-layer telemetry with anomaly detection, RCA, anomaly-duration, and forecasting tasks. RCAEval should provide the stronger root-cause and service-management evaluation because it contains annotated microservice failures and reproducible RCA baselines. SMD should be retained only for out-of-domain generalization; it is server telemetry, not telecom telemetry, and should not be described as a proxy for radio KPIs.

Positioning principle: do not try to claim all TNSM scope areas. Build a deep, experimentally defensible story around four anchors, use a fifth as supporting evidence, and avoid weak claims such as “O-RAN” unless the implementation or dataset explicitly uses O-RAN interfaces or components.

1. TNSM Scope Basis

TNSM publishes archival work advancing the state of the art and practical applications of network and service management. The scope supplied for this report groups the journal's interests into eight areas: Management Models, Architectures and Frameworks; Service Provisioning, Reliability and Quality Assurance; Management Functions; Enabling Technologies; Information and Communication Models; Policies; Applications and Case Studies; and Emerging Technologies and Standards. The current IEEE Communications Society journal page lists the same underlying topics and further emphasizes significant contributions in network/service management, including performance evaluation, scalability analysis, and optimization [1].

2. Recommended Primary TNSM Anchors

Anchor	Paper role	Evidence to include	Primary dataset
Management Functions	Core	Anomaly detection; fault localization; root-cause ranking; time-to-detect; top-k RCA accuracy.	TelecomTS + RCAEval
Service Provisioning, Reliability & QA	Core	Pre-degradation / early-warning experiments; robustness under faults; lead time; reliability-oriented evaluation.	TelecomTS + RCAEval
Enabling Technologies	Core	ML / graph / transformer / causal method, ablations, efficiency, confidence or abstention mechanism.	All datasets
Emerging Technologies & Standards	Core domain anchor	5G RAN observability and management using real telecom KPIs. Avoid claiming O-RAN unless directly supported.	TelecomTS
Applications & Case Studies	Supporting	Realistic 5G case study, failure scenarios, operational interpretation, and management usefulness.	TelecomTS

Why four primary anchors? A paper that superficially touches every scope item risks appearing unfocused. These four anchors form a coherent causal chain: network telemetry is observed, abnormal behavior is detected, degradation is anticipated or quantified, root causes are localized, and the resulting management action is supported by an enabling ML system. The fifth anchor - Applications and Case Studies - demonstrates that the method is not merely benchmark optimization.

3. Anchor-by-Anchor Research Positioning

3.1 Management Functions

This should be the paper's most explicit management contribution. Frame anomaly detection as a fault/performance-management function, not as generic time-series classification. The system should produce actionable outputs: anomaly score, affected KPI(s), likely root cause, confidence, and time-to-detection. RCAEval is especially valuable here because its failure cases include annotated root-cause services and indicators, while TelecomTS directly supports anomaly detection and root-cause analysis tasks [2][3].

3.2 Service Provisioning, Reliability and Quality Assurance

The strongest claim is proactive degradation assessment rather than only post-failure detection. TelecomTS provides raw scenario telemetry and forecasting support, but the dataset alone does not prove that a detectable precursor exists before every anomaly. Therefore, early-warning must be demonstrated experimentally. Evaluate multiple prediction horizons (for example 1 s, 5 s, 10 s, and 30 s before an event) and report lead time versus precision/recall. If predictive signal disappears before the event, narrow the claim to rapid detection and reliability assessment rather than “failure prediction.”

3.3 Enabling Technologies

The ML method belongs here, but it should serve the management objective. Whether the final architecture is a transformer, temporal graph model, causal model, or hybrid, the paper should show why that mechanism is needed for telecom observability: high-dimensional cross-KPI dependency, burstiness, heterogeneous covariates, non-stationarity, and the need for interpretable or confidence-aware decisions. Include ablations, parameter sensitivity, runtime, memory footprint, and scalability so the enabling technology is evaluated as an operational component rather than as an isolated model.

3.4 Emerging Technologies and Standards

TelecomTS gives a strong 5G emerging-technology anchor because it is derived from a 5G telecommunications testbed and contains high-resolution PHY, MAC, and network-layer KPIs sampled at 10 Hz [2][3]. Use terminology such as “5G RAN observability” or “5G network management.” Do not automatically label the work O-RAN: radio metrics such as RSRP, BLER, SNR, PRB utilization, and MCS are consistent with RAN observability but do not by themselves establish the use of O-RAN SMO/O1/E2/RIC interfaces.

3.5 Applications and Case Studies (Supporting)

Use this area to convert the benchmark into an operational story. Present representative anomalies, show which KPIs moved first, show the model’s warning/detection and RCA output, and explain the management consequence. A case-study section can make the difference between “model accuracy” and “network management usefulness,” particularly if it includes failure timelines, confidence/abstention behavior, and discussion of false alarms.

4. Dataset Strategy

Dataset	Domain role	Strongest contribution	Key limitation	Recommended use
TelecomTS	Primary telecom / 5G domain benchmark	5G testbed; 32k samples; 1M+ observations; PHY/MAC/network KPIs; AD, RCA, duration, forecasting tasks [2][3].	Processed samples are short windows; early-warning signal must still be proven. O-RAN is not established by the dataset alone.	Main experiments; 5G scope anchor; anomaly detection; early warning; telecom RCA.
RCAEval	Service-management / RCA rigor benchmark	Nine datasets, 735 failure cases, 11 fault types, annotated root causes, and reproducible RCA baselines across metrics/logs/traces [4].	Microservice systems rather than RAN; does not provide the 5G emerging-technology anchor.	RCA comparison; root-cause ranking; systems rigor; multi-source evaluation.

Dataset	Domain role	Strongest contribution	Key limitation	Recommended use
SMD	Out-of-domain generalization benchmark	28 server-machine entities with 38 dimensions; established multivariate anomaly benchmark [5].	Generic server telemetry, not telecom; no 5G or service-topology RCA semantics.	Generalization / robustness only; avoid presenting as telecom proxy.

Recommended dataset sentence for the manuscript: “TelecomTS serves as the primary domain benchmark for high-resolution 5G RAN observability; RCAEval evaluates root-cause localization under diverse service failures and established RCA baselines; and SMD provides an out-of-domain test of generalization across heterogeneous server telemetry.”

5. Proposed Evaluation Architecture

TelecomTS	RCAEval	SMD
5G RAN domain validity	Service-management rigor	Out-of-domain robustness
AD + early warning + telecom RCA	Root-cause localization + baseline comparison	Generalization across server telemetry

The method should be identical across datasets wherever possible. Dataset-specific preprocessing is acceptable, but changing the core algorithm for each benchmark weakens the generalization argument. The evaluation should clearly separate domain-specific success (TelecomTS), RCA-system rigor (RCAEval), and cross-domain robustness (SMD).

6. Minimum Experiment Set for a TNSM-Oriented Submission

Experiment	Dataset(s)	Primary metrics	TNSM purpose
Point / window anomaly detection	TelecomTS, SMD	AUROC, AUPRC, F1, event-based precision/recall, false alarms	Management Functions
Early-warning / degradation forecasting	TelecomTS	Lead time, recall at horizon, precision, false-alarm rate, horizon-vs-performance curve	Reliability & QA
Root-cause localization	TelecomTS, RCAEval	Top-1/Top-3/Top-5 accuracy, MRR, Avg@K where applicable	Management Functions
Baseline comparison	RCAEval + TelecomTS	Task metrics versus strong reproducible baselines	Systems evaluation rigor

Experiment	Dataset(s)	Primary metrics	TNSM purpose
Ablation and sensitivity	All	Performance delta, stability, calibration	Enabling Technologies
Efficiency and scalability	All / selected large subsets	Latency, throughput, memory, parameter count, scaling with channels/nodes	Operational feasibility
Cross-domain generalization	SMD	Performance retention / degradation versus in-domain results	Generalization evidence
Case-study timeline	TelecomTS	Qualitative + quantitative event timeline	Applications & Case Studies

7. Recommended Research Questions

- **RQ1 - Detection:** Can the proposed method detect anomalous 5G network behavior more accurately and with fewer false alarms than representative time-series baselines?
- **RQ2 - Anticipation:** How early can the method identify impending degradation while maintaining operationally acceptable precision and false-alarm rates?
- **RQ3 - Root Cause:** Can the method localize the responsible KPI/service/component under heterogeneous network and service failures?
- **RQ4 - Robustness:** How stable is performance across anomaly types, telemetry modalities, channel counts, and out-of-domain server telemetry?
- **RQ5 - Operationality:** What latency, throughput, memory, and confidence-calibration characteristics determine whether the method is suitable for an online management loop?

8. Claims to Make

Use / defend	Avoid unless directly demonstrated
"5G RAN observability and management."	"O-RAN management" without O-RAN interfaces/components.
"Early-warning evaluation at multiple horizons."	"Predicts failures before disruption" unless pre-event signal is empirically demonstrated.
"SMD evaluates out-of-domain generalization."	"SMD is a telecom KPI proxy."
"RCAEval provides service-failure RCA rigor and reproducible baselines."	Implying RCAEval itself is a radio/network dataset.
"Management contribution: detection, diagnosis, confidence-aware decision support."	A paper whose only contribution is higher F1/AUROC on generic anomaly datasets.

9. Paper Structure

1. **I. Introduction:** 5G management problem, operational consequences, research gap, contributions.
2. **II. Background and Related Work:** 5G observability, anomaly detection, RCA, proactive reliability, management context.
3. **III. Problem Formulation:** Telemetry, event/fault labels, prediction horizons, RCA target, confidence/abstention if used.
4. **IV. Proposed Management Method:** Architecture, temporal/cross-KPI modeling, RCA mechanism, complexity.
5. **V. Datasets and Experimental Protocol:** TelecomTS primary; RCAEval secondary; SMD generalization; splits and leakage controls.
6. **VI. Results:** Detection, early warning, RCA, ablation, efficiency/scalability, cross-domain generalization.
7. **VII. 5G Case Study and Management Implications:** Timeline-based telecom case study, failure interpretation, operational limits.
8. **VIII. Limitations and Threats to Validity:** Synthetic versus real anomalies, O-RAN limitation, precursor availability, benchmark transfer.
9. **IX. Conclusion:** Network/service-management contribution and future deployment path.

10. Final Decisions

Primary decision: keep TelecomTS as the scope anchor. It provides the strongest direct connection to 5G network observability and therefore protects the work from being read as generic server anomaly detection. Use RCAEval to establish stronger root-cause and service-management rigor, including comparisons against reproducible RCA baselines. Use SMD only to demonstrate that the method does not collapse outside the telecom domain.

The decisive experiment is no longer dataset "stitching"; it is predictive validity before anomaly onset. Measure whether pre-event windows contain enough information for meaningful warning at multiple horizons. If they do, the paper can defend a proactive reliability/quality-assurance contribution. If they do not, reposition that component as fast detection and reliability assessment rather than overstating prediction.

Best overall TNSM story: 5G RAN observability -> anomaly detection -> anticipatory degradation assessment -> root-cause localization -> confidence-aware management decision, evaluated with TelecomTS for domain validity, RCAEval for RCA rigor, and SMD for cross-domain robustness.

References

- [1] IEEE Communications Society, "IEEE Transactions on Network and Service Management," journal information and scope. <https://www.comsoc.org/publications/journals/ieee-transactions-network-and-service-management>
- [2] A. Feng et al., "TelecomTS: A Multi-Modal Observability Dataset for Time Series and Language Analysis," 2025, arXiv:2510.06063; dataset repository. <https://github.com/Ali-maatouk/TelecomTS>
- [3] AliMaatouk, "TelecomTS" dataset card, Hugging Face. <https://huggingface.co/datasets/AliMaatouk/TelecomTS>
- [4] Q. Pham et al., "RCAEval: A Benchmark for Root Cause Analysis," open-source benchmark repository. <https://github.com/phamquilluan/RCAEval>
- [5] NetManAIOps, "OmniAnomaly" repository and Server Machine Dataset (SMD) statistics. <https://github.com/NetManAIOps/OmniAnomaly>

Source note: The TNSM scope grouping and requested four-anchor strategy are based on the scope text and research framing supplied for this report. Dataset facts are cross-checked against the public project repositories/dataset cards cited above. Proposed experiments, claims discipline, and paper structure are research-design recommendations rather than claims made by those sources.